

Principal Components Analysis

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Linear Algebra

Homogeneous Linear Equations

- $A \iff B$: *An integer is even (A) if and only if an integer is divisible by 2 (B).*
- $\mathbf{Ax} = \mathbf{0}$ has a non-trivial solution if and only if columns of $\mathbf{A}$ are linearly dependent.
- $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution if and only if columns of $\mathbf{A}$ are linearly independent.
- Does $\mathbf{A}$ need to be a square matrix ?

$$\mathbf{Ax} = \mathbf{0}$$

$$\Rightarrow [\mathbf{a}_1 \ \mathbf{a}_2 \ \dots \ \mathbf{a}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\Rightarrow x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \dots + x_n \mathbf{a}_n = \mathbf{0}$$

Determinants

- Under row operations, a zero determinant cannot become non-zero and vice-versa.
- Columns of $\mathbf{A}$ linearly independent: By definition of linear independence, the only possible solution to $\mathbf{Ax} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$ i.e. $I_{n \times n} \mathbf{x} = \mathbf{0}$. $\Rightarrow$ The determinant of $I_{n \times n}$ is non-zero $\Rightarrow$ The determinant of $\mathbf{A}$ is non-zero and cannot be zero under any circumstances.
- Columns of $\mathbf{A}$ are linearly dependent $\Rightarrow$ One row is a linear combination of some of the other rows (row rank = column rank) $\Rightarrow$ Row operations can create a row of zeros.

Homogeneous Linear Equations: Square Matrices

- Consider a square matrix $\mathbf{A}$.
- We denote the determinant of a square matrix $\mathbf{A}$ by $|\mathbf{A}|$.
- $|\mathbf{A}| = 0$ if and only if $\mathbf{A}$ has linearly dependent columns.
- $|\mathbf{A}| \neq 0$ if and only if $\mathbf{A}$ has linearly independent columns.
- The system of linear equations $\mathbf{Ax} = \mathbf{0}$ has non-trivial solutions if and only if $|\mathbf{A}| = 0$.

Rank-Nullity Theorem

- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. The *nullity* of $\mathbf{A}$ is the dimension of its null space, that is, $\text{nullity}(\mathbf{A}) = \dim\{\mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0}\}$.
- Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be a matrix. Then $\text{nullity}(\mathbf{A}) + \text{rank}(\mathbf{A}) = n$.

Eigenvalues and Eigenvectors

- Suppose $\mathbf{Ax} = \lambda\mathbf{x}$. Here, $\mathbf{x}$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ provided $\mathbf{x} \neq \mathbf{0}$. λ can equal 0.
- The relation $\mathbf{Ax} = \lambda\mathbf{x}$ requires $\mathbf{A}$ to be a square matrix. $\mathbf{A}$ is $n \times n$ when $\mathbf{x}$ is $n \times 1$.
- $\mathbf{Ax} = \lambda\mathbf{x} \Rightarrow (\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$
- The determinant $|\mathbf{A} - \lambda\mathbf{I}|$ is the characteristic polynomial of the matrix $\mathbf{A}$.
- For a non-trivial solution $\mathbf{x}$ to exist for $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$, we must have $|\mathbf{A} - \lambda\mathbf{I}| = 0$.
- Eigenvalues are the roots of the characteristic polynomial.

Geometric and Algebraic multiplicity

- Let $\mathbf{A} \in \mathbb{F}^{n \times n}$ and let λ be an eigenvalue of $\mathbf{A}$. The *eigenspace* corresponding to λ is $E_\lambda = \{ \mathbf{x} \in \mathbb{F}^n : \mathbf{A}\mathbf{x} = \lambda\mathbf{x} \}$.
- The number of times a particular eigenvalue appears as a root of the characteristic polynomial is the algebraic multiplicity of the eigenvalue.
- The dimension of the eigenspace of an eigenvalue is the geometric multiplicity of the eigenvalue.
- Geometric multiplicity $\leq$ Algebraic multiplicity for each distinct eigenvalue.
- The dimension of the eigenspace of an eigenvalue λ is the nullity of $(\mathbf{A} - \lambda\mathbf{I})$.

Symmetric Matrices

- Real Symmetric Matrix: All eigenvalues are real
- Real Symmetric Matrix: Eigenvectors of distinct eigenvalues are orthogonal

Theorems

- A matrix $\mathbf{A} \in \mathbb{F}^{n \times n}$ is said to be *diagonalizable* if there exists an invertible matrix $\mathbf{P} \in \mathbb{F}^{n \times n}$ and a diagonal matrix $\mathbf{D} \in \mathbb{F}^{n \times n}$ such that $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$
- Theorem 1: A square matrix $\mathbf{P}$ is orthogonal if and only if $\mathbf{P}^T = \mathbf{P}^{-1}$.
- Theorem 2: Spectral Theorem: A real symmetric matrix is orthogonally diagonalizable.
- A square matrix is diagonalizable if and only if the geometric multiplicity of each eigenvalue equals the algebraic multiplicity.
- From Theorem 1 and Theorem 2, the geometric multiplicity of each eigenvalue equals the algebraic multiplicity for any real symmetric matrix.

Example

- $\mathbf{A} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$
- $\lambda = 1, 1, 4$
- For $\lambda = 1$, algebraic multiplicity: 2, geometric multiplicity: 2.
- Basis vectors for eigenspace: $\frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \frac{1}{\sqrt{3/2}} \begin{bmatrix} -1/2 \\ -1/2 \\ 1 \end{bmatrix}$
- For $\lambda = 4$, algebraic multiplicity: 1, geometric multiplicity: 1.
- Basis vector for eigenspace: $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Construct a set of orthonormal eigenvectors for a $n \times n$ real symmetric matrix $\mathbf{A}$

- The spectral theorem asserts that any real symmetric matrix $\mathbf{A}$ is diagonalizable.
- Consider a $n \times n$ real symmetric matrix $\mathbf{A}$.
- The characteristic polynomial is an n^{th} order polynomial.
- An n^{th} order polynomial has n roots.
- The sum of algebraic multiplicities is equal to n .
- Since the matrix $\mathbf{A}$ is diagonalizable, the geometric multiplicity equals the algebraic multiplicity for each eigenvalue.
- The sum of geometric multiplicities is therefore also equal to n .
- Since the matrix $\mathbf{A}$ is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal to each other. In other words, the eigenspaces are mutually orthogonal.

Construct a set of orthonormal eigenvectors for a $n \times n$ real symmetric matrix **A**

- Let there be p distinct eigenvalues: $\lambda_1, \lambda_2, \dots, \lambda_p$, $p \leq n$.
- Corresponding eigenspaces: $E_{\lambda_1}, E_{\lambda_2}, \dots, E_{\lambda_p}$
- $\dim(E_{\lambda_i}) = n_i$
- $\sum_{i=1}^n n_i = n_1 + n_2 + \dots + n_p = n$ (Sum of geometric multiplicities is n)

Construct a set of orthonormal eigenvectors for a $n \times n$ real symmetric matrix **A**

- Construct a set of orthonormal eigenvectors for each eigenspace: $\{\mathbf{v}_{\lambda_1}^1, \mathbf{v}_{\lambda_1}^2, \dots, \mathbf{v}_{\lambda_1}^{n_1}\}, \{\mathbf{v}_{\lambda_2}^1, \mathbf{v}_{\lambda_2}^2, \dots, \mathbf{v}_{\lambda_2}^{n_2}\}, \dots$
- $(\mathbf{v}_{\lambda_j}^i) \cdot \mathbf{v}_{\lambda_l}^k = 0, j \neq l$ (Eigenvectors belonging to distinct eigenspaces are orthogonal as they correspond to distinct eigenvalues)
- $(\mathbf{v}_{\lambda_j}^i) \cdot \mathbf{v}_{\lambda_j}^k = 0, k \neq i$ (Distinct eigenvectors belonging to the same eigenspace are orthogonal)
- The entire collection of eigenvectors is an orthonormal set.
- Since there are a total of n eigenvectors in $\mathbb{R}^n$, we have constructed an orthonormal basis for $\mathbb{R}^n$.
- We denote the set of eigenvectors as $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

Optimization Problem

- We restrict the search space to unit vectors $\mathbf{x}$ i.e. $\|\mathbf{x}\| = 1$.

$$\begin{array}{ll} \text{Maximize} & \|\mathbf{Ax}\|^2 \\ \text{such that} & \|\mathbf{x}\| = 1 \end{array} \quad (1)$$

Optimization Problem

- A real symmetric matrix $A \in \mathbb{R}^{n \times n}$ is called *positive semidefinite* if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$.
- $(\mathbf{A}^T \mathbf{A})^T = \mathbf{A}^T (\mathbf{A}^T)^T = \mathbf{A}^T \mathbf{A}$
- $\mathbf{x}^T (\mathbf{A}^T \mathbf{A}) \mathbf{x} = (\mathbf{x}^T \mathbf{A}^T)(\mathbf{A} \mathbf{x}) = (\mathbf{A} \mathbf{x})^T (\mathbf{A} \mathbf{x}) = \|\mathbf{A} \mathbf{x}\|^2 \geq 0$
- $\mathbf{A}^T \mathbf{A}$ is symmetric and positive semidefinite: Eigenvalues are non-negative
- Since $\mathbf{A}^T \mathbf{A}$ is symmetric, one can construct an orthonormal set of eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^n$:
 $\mathbf{A}^T \mathbf{A} \mathbf{v}_i = \lambda_i \mathbf{v}_i = \sigma_i^2 \mathbf{v}_i$
- $\sigma_i = \sqrt{\lambda_i}$ are the singular values of $\mathbf{A}$
- $\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$
- $\mathbf{v}_i^T \mathbf{v}_i = \|\mathbf{v}_i\|^2 = 1, \mathbf{v}_i^T \mathbf{v}_j = 0$
- $\mathbf{x}^T \mathbf{x} = 1 \Rightarrow c_1^2 + c_2^2 + \dots + c_n^2 = 1.$

Optimization Problem

$$\begin{aligned}\|\mathbf{Ax}\|^2 &= (\mathbf{Ax})^T \mathbf{Ax} = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} \\ \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} &= \mathbf{x}^T \mathbf{A}^T \mathbf{A}(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n) \\ &= \mathbf{x}^T (c_1 \mathbf{A}^T \mathbf{A} \mathbf{v}_1 + c_2 \mathbf{A}^T \mathbf{A} \mathbf{v}_2 + \dots + c_n \mathbf{A}^T \mathbf{A} \mathbf{v}_n) \\ &= \mathbf{x}^T (c_1 \lambda_1 \mathbf{v}_1 + c_2 \lambda_2 \mathbf{v}_2 + \dots + c_n \lambda_n \mathbf{v}_n) \\ &= (c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n)^T (c_1 \lambda_1 \mathbf{v}_1 + \dots + c_n \lambda_n \mathbf{v}_n) \\ &= c_1^2 \lambda_1 + \dots + c_n^2 \lambda_n \\ &= c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2\end{aligned}\tag{2}$$

Optimization Problem

- Arrange singular values as $\sigma_1^2 \geq \sigma_2^2 \geq \dots \geq \sigma_n^2$
- Corresponding eigenvectors: $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$
- $\|\mathbf{Ax}\|^2 = c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2 \leq c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_1^2 = \sigma_1^2 (c_1^2 + \dots + c_n^2) = \sigma_1^2$
- Select $\mathbf{x} = \mathbf{v}_1$
- $\mathbf{x} = \mathbf{v}_1 \Rightarrow c_1 = 1, c_2 = c_3 = \dots = c_n = 0$
- $\|\mathbf{Ax}\|^2 = c_1^2 \sigma_1^2 + \dots + c_n^2 \sigma_n^2 = \sigma_1^2$
- The solution to the optimization problem is obtained by selecting $\mathbf{x} = \mathbf{v}_1$, i.e. the eigenvector corresponding to the largest eigenvalue.
- If the search space is constrained to the subspace orthogonal to $\mathbf{v}_1$, then $\{\mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis in this subspace.
- The solution to the optimization problem is obtained by selecting $\mathbf{x} = \mathbf{v}_2$, i.e. the eigenvector corresponding to the second largest eigenvalue.
- One can similarly extend to other eigenvectors.

Principal Components Analysis

Data Matrix

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix}$$

- $\mathbf{X}$ is the data matrix
- There are n samples $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$
- There are p features

- $\mathbf{x}_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{ip} \end{bmatrix} \Rightarrow \mathbf{x}_i^T = [x_{i1} \ x_{i2} \ \cdots \ x_{ip}]$

Centering Matrix

$$\begin{aligned}\mathbf{H} &= \mathbf{I}_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \\ &= \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} - \frac{1}{n} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} - \frac{1}{n} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix}\end{aligned}$$

Centred Data Matrix

$$\mathbf{X}_c = \mathbf{H}\mathbf{X} = \begin{bmatrix} x_{11} - \frac{1}{n} \sum_{i=1}^n x_{i1} & x_{12} - \frac{1}{n} \sum_{i=1}^n x_{i2} & \cdots & x_{1p} - \frac{1}{n} \sum_{i=1}^n x_{ip} \\ x_{21} - \frac{1}{n} \sum_{i=1}^n x_{i1} & x_{22} - \frac{1}{n} \sum_{i=1}^n x_{i2} & \cdots & x_{2p} - \frac{1}{n} \sum_{i=1}^n x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \frac{1}{n} \sum_{i=1}^n x_{i1} & x_{n2} - \frac{1}{n} \sum_{i=1}^n x_{i2} & \cdots & x_{np} - \frac{1}{n} \sum_{i=1}^n x_{ip} \end{bmatrix}$$

$$\mathbf{X}_c = \begin{bmatrix} \hat{\mathbf{x}}_1^T \\ \hat{\mathbf{x}}_2^T \\ \vdots \\ \hat{\mathbf{x}}_n^T \end{bmatrix}$$

Centred Data Matrix

- $\mathbf{X}_c$ is the centered data matrix.
- $\mathbf{X}_c$ is $n \times p$.
- Each column of $\mathbf{X}_c$ adds to zero
- Equivalently, the sample mean is the null vector:

$$\bar{\mathbf{x}}^T = \frac{1}{n} \left(\hat{\mathbf{x}}_1^T + \hat{\mathbf{x}}_2^T + \dots \hat{\mathbf{x}}_1^T \right) = \mathbf{0}_{1 \times p}$$

Example: Data Matrix Centering

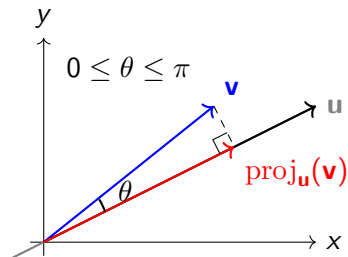
$$\mathbf{X} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 1 \\ 4 & 3 \end{bmatrix}$$

$$\bar{\mathbf{x}} = \begin{bmatrix} \frac{1+2+3+4}{4} \\ \frac{1+3+1+3}{4} \end{bmatrix} = \begin{bmatrix} 2.5 \\ 2 \end{bmatrix}$$

$$\mathbf{X}_c = \begin{bmatrix} -1.5 & -1 \\ -0.5 & 1 \\ 0.5 & -1 \\ 1.5 & 1 \end{bmatrix}$$

$$\begin{aligned} \hat{\mathbf{x}}^T &= [(-1.5 - 0.5 + 0.5 + 1.5) \quad (-1 + 1 - 1 + 1)] \\ &= [0 \quad 0] = \mathbf{0}_{1 \times 2} \end{aligned}$$

Projection of a vector $\mathbf{v}$ onto another vector $\mathbf{u}$



$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \|\mathbf{v}\| \cos(\theta) \hat{\mathbf{u}} \quad (\| \text{ refers to Euclidean norm})$$

$$[\text{Signed magnitude } \text{proj}_{\mathbf{u}}(\mathbf{v}) : \|\mathbf{v}\| \cos(\theta)]$$

$$= \frac{\|\mathbf{v}\| \cos(\theta)}{\|\mathbf{u}\|} \mathbf{u}$$

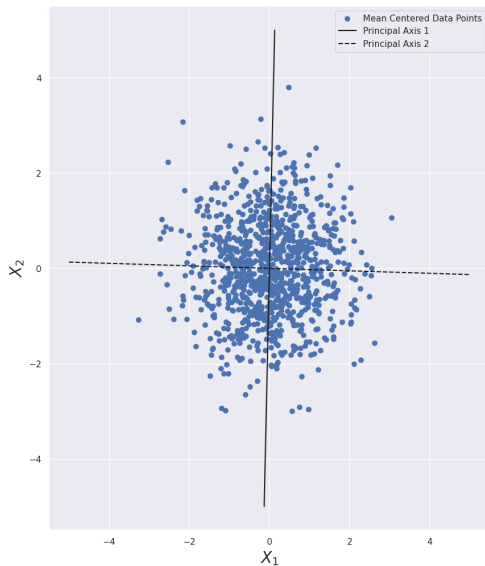
$$= \frac{\|\mathbf{v}\| \|\mathbf{u}\| \cos(\theta)}{\|\mathbf{u}\|^2} \mathbf{u}$$

$$= \frac{\mathbf{v} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

Signed Magnitude of Projection

$$\begin{aligned}\text{Signed Magnitude } \text{proj}_{\mathbf{u}}(\mathbf{v}) &= \|\mathbf{v}\| \cos(\theta) \\ &= \|\mathbf{v}\| \|\mathbf{u}\| \cos(\theta) \text{ (assuming } \|\mathbf{u}\| = 1) \\ &= \mathbf{v}^T \mathbf{u} = \mathbf{v} \cdot \mathbf{u}\end{aligned}$$

Principal Components



Variable Transformation

$$\hat{\mathbf{x}} = [\hat{x}_1 \quad \hat{x}_2 \quad \hat{x}_3 \quad \hat{x}_4]$$

$$\mathbf{u} = \frac{1}{\sqrt{10}} [1 \quad 2 \quad 2 \quad 1]$$

$$\hat{\mathbf{x}} \cdot \mathbf{u} = \frac{1}{\sqrt{10}} (\hat{x}_1 + 2\hat{x}_2 + 2\hat{x}_3 + \hat{x}_4)$$

Variable Transformation

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \mathbf{X}_c \mathbf{u} = \begin{bmatrix} \hat{\mathbf{x}}_1^T \\ \hat{\mathbf{x}}_2^T \\ \vdots \\ \hat{\mathbf{x}}_n^T \end{bmatrix} \mathbf{u} = \begin{bmatrix} \hat{\mathbf{x}}_1^T \mathbf{u} \\ \hat{\mathbf{x}}_2^T \mathbf{u} \\ \vdots \\ \hat{\mathbf{x}}_n^T \mathbf{u} \end{bmatrix}$$

Variable Transformation

$$y_i = \hat{\mathbf{x}}_i^T \mathbf{u}$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

$$= \frac{1}{n} \sum_{i=1}^n \hat{\mathbf{x}}_i^T \mathbf{u}$$

$$= \frac{1}{n} \left(\sum_{i=1}^n \hat{\mathbf{x}}_i^T \right) \mathbf{u}$$

$$= \frac{1}{n} \mathbf{0}_{1 \times p} \mathbf{u}_{p \times 1}$$

$$= 0$$

Variable Transformation

$$\begin{aligned}\sigma_y^2 &= \frac{1}{n-1} \sum_{i=1}^{n-1} (y_i - \bar{y})^2 \\&= \frac{1}{n-1} \sum_{i=1}^n y_i^2 \\&= \frac{1}{n-1} \|\mathbf{y}\|^2 \\&= \frac{1}{n-1} \|\mathbf{X}_c \mathbf{u}\|^2 \\&= \frac{1}{n-1} (\mathbf{X}_c \mathbf{u})^T (\mathbf{X}_c \mathbf{u}) \\&= \frac{1}{n-1} \mathbf{u}^T \mathbf{X}_c^T \mathbf{X}_c \mathbf{u} \\&= \mathbf{u}^T \left(\frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c \right) \mathbf{u} = \mathbf{u}^T \mathbf{S} \mathbf{u}\end{aligned}$$

Variable Transformation: Key results

$$\mathbf{y} = \mathbf{X}_c \mathbf{u}$$

$$\sigma_y^2 = \frac{1}{n-1} \|\mathbf{X}_c \mathbf{u}\|^2$$

$$\sigma_y^2 = \mathbf{u}^T \left(\frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c \right) \mathbf{u} = \mathbf{u}^T \mathbf{S} \mathbf{u}$$

PCA Objectives

- $\mathbf{u}_1 = \arg \max_{\mathbf{u}} \mathbf{u}^\top \mathbf{S} \mathbf{u}$ subject to $\|\mathbf{u}\| = 1$
- $\mathbf{u}_2 = \arg \max_{\mathbf{u}} \mathbf{u}^\top \mathbf{S} \mathbf{u}$ subject to $\|\mathbf{u}\| = 1, \quad \mathbf{u}^\top \mathbf{u}_1 = 0$
- $\mathbf{u}_k = \arg \max_{\mathbf{u}} \mathbf{u}^\top \mathbf{S} \mathbf{u}$ subject to $\|\mathbf{u}\| = 1, \quad \mathbf{u}^\top \mathbf{u}_j = 0 \quad \forall j < k$

Eigenvalues and Eigenvectors of the sample variance-covariance matrix

$$\mathbf{X}_c^T \mathbf{X}_c \mathbf{v}_i = \lambda_i \mathbf{v}_i$$

$$\left(\frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c \right) \mathbf{v}_i = \frac{1}{n-1} \lambda_i \mathbf{v}_i$$

$$\mathbf{S} \mathbf{v}_i = \left(\frac{1}{n-1} \lambda_i \right) \mathbf{v}_i$$

$$\mathbf{S} \mathbf{v}_i = \lambda_i^S \mathbf{v}_i = (\sigma_i^S)^2 \mathbf{v}_i, \text{ where } \sigma_i^S = \sqrt{\lambda_i^S}$$

Solution to the PCA problem

- Order eigenvalues from largest to smallest:
 $\lambda_1^S \geq \lambda_2^S \geq \dots \geq \lambda_n^S$
- Corresponding eigenvectors: $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ ($\mathbf{v}_1$ is the eigenvector corresponding to the largest eigenvalue, $\mathbf{v}_2$ is the eigenvector corresponding to the second largest eigenvalue, and so on)
- The first principal component is the eigenvector corresponding to the largest eigenvalue, i.e. $\mathbf{v}_1$.
- The second principal component is the eigenvector corresponding to the second largest eigenvalue, i.e. $\mathbf{v}_2$.
- Other principal components are obtained similarly.

First Principal Component: Objective Function Value

$$\begin{aligned}\max_{\mathbf{u}} \mathbf{u}^T \mathbf{S} \mathbf{u} &= \frac{1}{n-1} \|\mathbf{X}_c \mathbf{v}_1\|^2 \\&= \frac{1}{n-1} \mathbf{v}_1^T \mathbf{X}_c^T \mathbf{X}_c \mathbf{v}_1 \\&= \mathbf{v}_1^T \left(\frac{1}{n-1} \mathbf{X}_c^T \mathbf{X}_c \mathbf{v}_1 \right) \\&= \mathbf{v}_1^T (\mathbf{S} \mathbf{v}_1) \\&= \mathbf{v}_1^T (\lambda_1^S \mathbf{v}_1) \\&= \lambda_1^S \mathbf{v}_1^T \mathbf{v}_1 = \lambda_1^S \|\mathbf{v}_1\|^2 = \lambda_1^S\end{aligned}$$

Explained Variance

- Variance explained by the i^{th} principal component: λ_i^S .
- Total variance in the data = $\sum_{i=1}^n \lambda_i^S = \text{tr}(\mathbf{S})$.
- Fractional variance explained by the i^{th} principal component:
$$\frac{\lambda_i^S}{\sum_{i=1}^n \lambda_i^S}$$

PCA Scores Matrix

- $\mathbf{X}_c \in \mathbb{R}^{n \times p}$
- $\mathbf{U} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_k] \in \mathbb{R}^{p \times k}$
- $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ are the first k principal components
- $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k$
- Scores Matrix: $\mathbf{T} = \mathbf{X}_c \mathbf{U}$
- $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_k$